

Table 11.2 Regression Results

	$\ln(a_1)$	a_2	a_3	a_4
Estimate	6.66	0.57	0.78	0.52
Std Error	0.15	0.02	0.09	0.03
t-stat	43.88	34.59	8.76	16.34
R ² :	0.93			
SeY:	0.20			
F-Stat.	513.47			
Calculation				
$\ln(a_1)$	6.66			
seY	0.020697			
a_1^*	6.68			
a_1	793.3635			

volatility, and strategy (in terms of percentage of volume, time, or trading rate), users will find that the simplified I-Star model works extremely well in the pre-trade of pre-trade process.

Therefore, the simplified I-Star model is:

$$MI_{bp} = 793 \cdot Size^{0.57} \cdot \sigma^{0.78} \cdot POV^{0.52} \quad (11.4)$$

Recall that $E[\exp\{\ln(a_1)\}] = a_1 + \frac{1}{2} \cdot \sigma^2$.

Investors can also estimate the parameters of the full I-Star model including the b_1 parameter to differentiate between permanent and temporary impact cost if desired. This technique is also fairly straight-forward but requires more sophisticated estimation process such as non-linear least squares or maximum likelihood estimation since the equation cannot be linearized via log-transformation.

This pre-trade of pre-trade technique (also referred to as the aggregated model) allows investors to construct models that are a general consensus of market participants. It is a way to survey the market and draw a general conclusion of expectations. The most appealing aspect of using a pre-trade of pre-trade model is that it provides investors and analysts with a quick and easy process to generate the necessary cost data to fit these models. Investors and buy-side analysts often have difficulty constructing the input data set for their in-house model since the process is extremely data intensive and resource draining.